

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
	(c)		Correct substitution into $v^2 = u^2 + 2ax$ (i.e. $0 = u^2 - (2 \times 8 \times 85)$ [1] don't award mark if incorrect signs $u = 36.9 \text{ [ms}^{-1}\text{]}$ ecf on incorrect signs [1] No, greater than speed limit [1] must correlate with working Alternative: Correct substitution into $v^2 = u^2 + 2ax$ using 30 ms^{-1} to find x. (i.e. $0 = (30)^2 - (2 \times 8 \times x)$ [1] don't award mark if incorrect signs $x = 56.3 \text{ [m]}$ ecf on incorrect signs [1] No, greater than speed limit [1] must correlate with working Alternative: Using $v = u + at$ to determine $t = 3.75 \text{ s}$ (1) Using $x = \frac{1}{2}(u + v)t$ i.e. $3.75 \times \text{mean velocity (15)} = 56.3 \text{ [m]}$ (1) No, greater than speed limit [1] must correlate with working			3	3	2	
			Question 2 total	3	4	3	10	7	0